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Electronic Medical Records (EMR)

Case Study

Item	Details
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1. Executive Summary

A small multi-doctor outpatient clinic currently keeps patient records on paper and in scattered spreadsheets. This makes it slow to find a patient's history, easy to lose records, and hard to produce reports for management or regulators. This case study defines the business need for an Electronic Medical Records (EMR) system, works through the requirements and process, and produces a prioritised, estimated and costed set of user stories that could be handed to a delivery team.

The process flow described in Section 7 covers the full patient journey: arrival, appointment check, EMR creation, consultation, diagnosis entry, payment, next-appointment scheduling, and reminder notifications.

2. Business Background and Problem Statement

The clinic assumed for this case study sees around 120 patients a day across 6 doctors. Today:

- Patient history is on paper folders, so a doctor cannot see a patient's past visits at another branch or with another doctor.
- Prescriptions are handwritten, which occasionally causes pharmacy errors from illegible writing.
- Lab results arrive on paper and are filed manually, so they are sometimes misplaced.
- Billing is done separately from the clinical record, so invoices sometimes miss a service that was given.
- Appointment reminders are manual, so patients miss follow-ups.
- Producing a report, for example how many patients were seen for a condition last month, takes days of manual counting.

Opportunity

An EMR system that holds one digital record per patient, used at registration, during the consultation, at billing, and for appointment scheduling, would remove most of these problems and let the clinic report on its own data.

3. Business Objectives

ID	Business objective	How it will be measured
BO-1	Give every doctor a complete, legible patient history at the point of care	Time to retrieve a patient's history (target: under 30 seconds)
BO-2	Reduce prescription and lab-filing errors caused by paper records	Reported prescription and filing errors per month
BO-3	Link billing directly to the services recorded during a visit	Share of invoices matching the visit record exactly
BO-4	Let management report on clinical and operational data without manual counting	Time to produce a monthly patient-volume report
BO-5	Reduce missed appointments through automatic reminders	No-show rate per month

4. Scope

4.1 In scope

- Patient arrival and appointment check
- Doctor availability and acceptance check
- Patient schedule check and reminder dispatch (email/SMS)
- Patient registration and a single digital record per patient (EMR form)
- Doctor consultation: history, examination notes, diagnosis
- Diagnosis entry into the EMR
- Billing generated from the visit record
- Payment recording
- Next-appointment scheduling
- Automatic email and SMS reminders for the next appointment
- Basic reports: patient volume, common diagnoses, revenue by service

4.2 Out of scope for this release

- Integration with external hospital or insurance systems
- A patient-facing mobile app or portal
- Inventory and pharmacy stock management
- Multi-branch, multi-currency billing
- Digital prescription printing and lab-result attachment (deferred to a later phase — see Section 9)

5. Stakeholders

Stakeholder	Interest
Doctors	Fast access to patient history, simple entry of diagnosis during consultation
Front-desk staff	Quick registration, appointment lookup and scheduling
Billing staff	Accurate, complete invoices and payment recording
Clinic management	Reliable reports on patient volume, diagnoses and revenue
Patients	Correct records, less waiting, timely reminders for next appointment

6. Business Requirements

ID	Business requirement	Objective
BR-01	The business needs one digital record per patient, viewable at any visit.	BO-1
BR-02	The business needs to check doctor availability and acceptance before assigning a patient.	BO-1
BR-03	The business needs to send appointment reminders by email and SMS.	BO-5
BR-04	The business needs billing generated directly from the recorded visit.	BO-3
BR-05	The business needs to record payments against invoices.	BO-3
BR-06	The business needs standard reports available without manual counting.	BO-4

7. Process Flow

The diagram below shows a patient visit from arrival to exit, across four lanes: Patient, Front desk, Clinician and the EMR system itself.

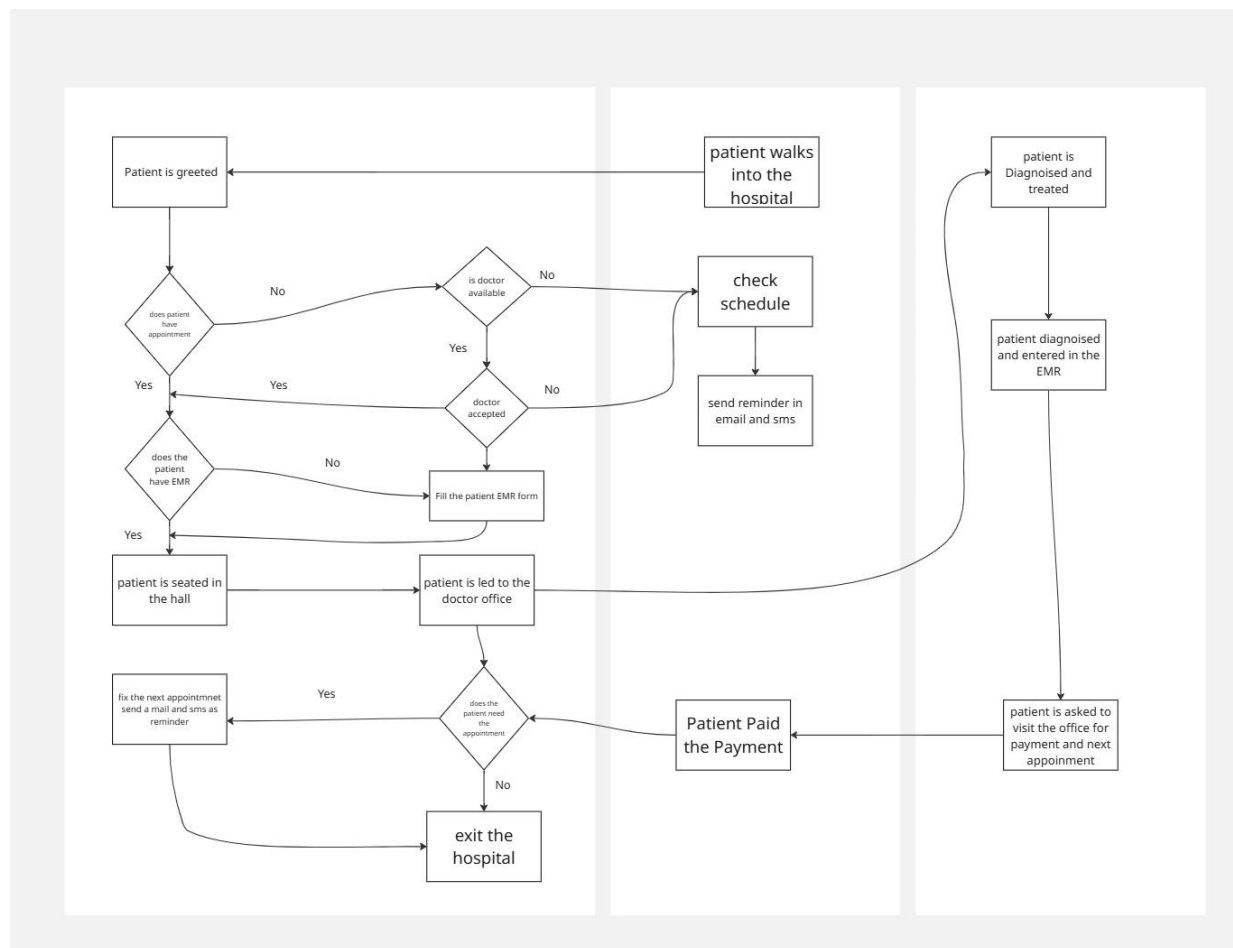


Figure 7: Process Flow Diagram

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8. User Stories

Priority uses MoSCoW: Must have, Should have, Could have.

ID	User story	Acceptance criteria	Priority	Links to
US-01	As front-desk staff, I want to greet and check in a patient, so that the visit begins.	Patient is marked as arrived in the system.	Must	BR-01
US-02	As front-desk staff, I want to check whether the patient has an appointment, so that I can route them correctly.	System shows appointment status within 2 seconds.	Must	BR-01
US-03	As front-desk staff, I want to check doctor availability and acceptance, so that I can assign the patient to an available doctor.	Doctor's availability and acceptance status is visible; patient is routed accordingly.	Must	BR-02
US-04	As front-desk staff, I want the system to send an email and SMS reminder when a doctor is unavailable, so that the patient is informed.	Email and SMS are dispatched within 1 minute of the decision.	Must	BR-03
US-05	As front-desk staff, I want to register a new patient and fill the EMR form, so that a record exists before the first visit.	A new record is created with name, date of birth, phone number and a unique patient ID.	Must	BR-01
US-06	As a doctor, I want to view the patient's EMR before consultation, so that I understand their history.	Past visits are listed newest first, each showing date, diagnosis and doctor seen.	Must	BR-01
US-07	As a doctor, I want to record examination notes and a diagnosis, so that the visit is documented in the EMR.	Notes and diagnosis are saved against the current visit and cannot be edited by anyone but that doctor.	Must	BR-01
US-08	As billing staff, I want an invoice generated from the visit record, so that I do not have to re-enter what was done.	The invoice lists the consultation and any ordered service, with no manual entry needed.	Must	BR-04
US-09	As billing staff, I want to record a payment against an invoice, so that the clinic's accounts are accurate.	A payment reduces the invoice balance and marks it paid when the balance reaches zero.	Must	BR-05
US-10	As front-desk staff, I want to schedule the next appointment, so that follow-up visits are planned.	Next appointment date and doctor are saved against the patient record.	Should	BR-03
US-11	As front-desk staff, I want the system to send an email and SMS reminder for the next appointment, so that the patient does not forget.	Reminder is dispatched at a configurable interval before the appointment.	Should	BR-03
US-12	As clinic management, I want a monthly patient-volume report, so that I can see how many patients were seen without counting manually.	A report shows patients seen per day and per doctor for a chosen month, exportable to Excel.	Should	BR-06

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ID	User story	Acceptance criteria	Priority	Links to
US-13	As clinic management, I want a report of the most common diagnoses, so that I can plan staffing and supplies.	A report ranks diagnoses by frequency for a chosen date range.	Could	BR-06

9. Estimation of User Stories

Estimation uses story points on the Fibonacci scale (1, 2, 3, 5, 8, 13). For budgeting, each point is converted to an illustrative 4 hours of effort.

ID	Story points	Size	Estimated effort (hours)	Notes
US-01	1	Small	4	Simple status flag
US-02	3	Small	12	Appointment lookup and routing logic
US-03	5	Medium	20	Availability and acceptance checks; routing rules
US-04	3	Small	12	Email + SMS integration
US-05	3	Small	12	New record with required fields and validation
US-06	5	Medium	20	History view linking past visits
US-07	5	Medium	20	Data entry screen with save and lock-after-doctor rules
US-08	8	Large	32	Invoice logic pulling services from the visit
US-09	5	Medium	20	Payment recording and balance calculation
US-10	3	Small	12	Appointment scheduling
US-11	3	Small	12	Reminder scheduling and dispatch
US-12	8	Large	32	Reporting engine, filters and Excel export
US-13	5	Medium	20	Second report reusing the reporting engine from US-12

Total: 57 story points, an estimated 228 hours across all 13 user stories.

10. Budgeting for Every User Story

Cost is estimated as effort (hours) × an illustrative blended rate of ₹800 per hour.

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ID	Estimated hours	Rate (illustrative)	Estimated cost	Priority
US-01	4	₹800/hr	₹3,200	Must
US-02	12	₹800/hr	₹9,600	Must
US-03	20	₹800/hr	₹16,000	Must
US-04	12	₹800/hr	₹9,600	Must
US-05	12	₹800/hr	₹9,600	Must
US-06	20	₹800/hr	₹16,000	Must
US-07	20	₹800/hr	₹16,000	Must
US-08	32	₹800/hr	₹25,600	Must
US-09	20	₹800/hr	₹16,000	Must
US-10	12	₹800/hr	₹9,600	Should
US-11	12	₹800/hr	₹9,600	Should
US-12	32	₹800/hr	₹25,600	Should
US-13	20	₹800/hr	₹16,000	Could

Budget summary by priority

Priority	Stories	Hours	Estimated cost
Must	US-01 to US-09	152	₹1,21,600
Should	US-10, US-11, US-12	56	₹44,800
Could	US-13	20	₹16,000
Total	13 stories	228	₹1,82,400

A phased delivery could build only the Must stories first (₹1,21,600, an estimated 152 hours), giving the clinic a working EMR for check-in, registration, consultation, diagnosis, invoicing and payment. Appointment scheduling, reminders and reporting would follow in a second phase.

11. Assumptions

- A single clinic location, around 120 patients a day across 6 doctors.
- A blended delivery rate of ₹800 per hour and a conversion of 4 hours per story point; both are illustrative.
- The clinic already has computers and internet access at the front desk and in consultation rooms.
- Email and SMS gateways are available and can be integrated.
- The process flow in Section 7 is treated as the authoritative source for scope in this version of the case study.
- Digital prescriptions and lab-result attachment (from the original draft) are deferred to a later phase, because they are not shown in the current process flow diagram.
- Estimates are at the size shown in Section 9; a real team would re-estimate after backlog refinement.

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12. Risks and Mitigations

Risk	Impact	Mitigation
Doctors resist moving away from paper notes	Slow adoption, records kept in two places	Involve doctors early; keep the consultation screen as fast as handwriting
Patient data exposure	Privacy breach, loss of trust	Role-based access; audit trail of who viewed or edited records
Incomplete migration of old paper records	Doctors cannot see full history for existing patients	Migrate the most recent 12 months of records first
Underestimated effort for invoicing logic (US-08)	Budget or timeline overrun	Treat US-08 as high-risk; review estimate after a technical design spike
Email/SMS gateway failure	Patients miss reminders	Retry logic and fallback to manual call list

13. Success Criteria

Measure	Target
Patient records available digitally at consultation	100% of registered patients
Invoices generated with zero missed services	100% of invoices
Time to produce a monthly patient-volume report	Same day, down from several days
Appointment reminders dispatched successfully	99% of scheduled reminders
No-show rate	Reduced by 30% within 6 months

14. Glossary

Term	Meaning
EMR	Electronic Medical Record: a digital patient record used within one clinic or provider
MoSCoW	Priority method: Must, Should, Could, Won't
Story point	A relative measure of effort and complexity for a user story
Velocity	How many story points a team completes per sprint, on average
Blended rate	A single hourly rate standing in for a mix of roles (analysis, build, test)

15. Approval

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Name	Role	Signature	Date
[Name]	Business analyst	[signature]	[Date]
[Name]	Reviewer	[signature]	[Date]